

Robust Halley’s Method for the Chord-Length Parameter L in Clothoid G^1 Hermite Interpolation

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Abstract

We present a numerically robust, cubic-convergent solver based on Halley’s method for the nonlinear chord-length residual formulation of the clothoid (Euler spiral) G^1 Hermite interpolation problem. The algorithm operates directly on the arc-length parameter L and employs fixed 32-point Gauss–Legendre quadrature over $[0, 1]$ for the six moment integrals that define the residual $f(L)$ and its first two analytic derivatives. Industrial-grade safety heuristics (derivative sign guards, denominator thresholds, and positivity damping) ensure reliable convergence even on challenging real-world railway alignment data from the ProRail Spoorgeometrie dataset.

We provide production-ready implementations in C# (.NET 8), Java 21, and TypeScript (Node.js), all producing bit-identical results. Benchmarks on approximately 180 real ProRail transition curves (“Overgangsboog”) demonstrate that Halley’s method consistently outperforms a comparable Newton variant, achieving convergence in an average of 5.3 iterations (vs. 7.8) with sub-4 μ s solve times. The mathematical derivations are cross-verified symbolically with SymPy and machine-checked in Coq using the Coquelicot real-analysis library, and the overall approach is situated within the current state-of-the-art, building upon the foundational work of Bertolazzi & Frego while offering an explicit L -formulation with enhanced robustness guarantees.

1 Introduction

Clothoids (also known as Euler spirals or Cornu spirals) are curves whose curvature varies linearly with arc length. They are the de-facto standard transition elements in railway and highway design because they provide G^2 continuity when connecting straight segments to circular arcs, ensuring smooth changes in lateral acceleration.

The G^1 Hermite interpolation problem with a single clothoid consists of finding parameters $L > 0$, κ_0 , and κ_1 (or equivalently the curvature rate) such that the clothoid segment starts at point P_0 with tangent angle θ_0 and curvature κ_0 , and ends at P_1 with tangent angle θ_1 and curvature κ_1 . In many practical settings—particularly when the endpoints and curvatures are known from design data or surveyed geometry—it is natural to solve directly for the arc length L .

This paper focuses on the *chord-length residual formulation* (L-formulation). We reduce the problem to a scalar nonlinear equation $f(L) = 0$ whose root is the desired arc length. We then apply Halley’s method, a cubic-convergent root-finding algorithm, augmented with carefully engineered safety heuristics that guarantee practical robustness.

Our contributions are:

- A complete, derivative-exact derivation of $f(L)$, $f'(L)$, and $f''(L)$ using six Gauss–Legendre moments.

- A production-grade Halley solver with explicit safeguards (sign checks, denominator guards, and damped updates).
- Cross-language implementations (C#, Java, TypeScript) with identical numerical behavior.
- Rigorous benchmarking on real ProRail railway transition data.
- Open academic documentation with full mathematical detail and SOTA comparison.

2 Mathematical Background

2.1 The Clothoid Parametrization

A clothoid is defined by the curvature function

$$\kappa(s) = \kappa_0 + \frac{\kappa_1 - \kappa_0}{L}s, \quad s \in [0, L],$$

where L is the arc length, $\kappa_0 = \kappa(0)$, and $\kappa_1 = \kappa(L)$. The heading (tangent angle) is obtained by integration:

$$\theta(s) = \theta_0 + \kappa_0 s + \frac{\kappa_1 - \kappa_0}{2L}s^2.$$

The Cartesian coordinates follow from the Fresnel integrals:

$$x(s) = x_0 + \int_0^s \cos \theta(u) du, \quad y(s) = y_0 + \int_0^s \sin \theta(u) du.$$

2.2 The Chord-Length Residual (L-Formulation)

Given P_0 , P_1 , κ_0 , and κ_1 , we seek $L > 0$ such that the chord length of the clothoid equals the Euclidean distance $d = \|P_1 - P_0\|$. Normalizing the parameter to $\tau = s/L \in [0, 1]$ yields the auxiliary function

$$\psi(\tau) = \kappa_0 \tau + \frac{\kappa_1 - \kappa_0}{2}\tau^2.$$

The six fundamental moments (evaluated by 32-point Gauss–Legendre quadrature on $[0, 1]$) are

$$P = \int_0^1 \cos(L\psi(\tau)) d\tau, \quad Q = \int_0^1 \sin(L\psi(\tau)) d\tau, \quad (1)$$

$$R = \int_0^1 \psi(\tau) \cos(L\psi(\tau)) d\tau, \quad T = \int_0^1 \psi(\tau) \sin(L\psi(\tau)) d\tau, \quad (2)$$

$$S_{2c} = \int_0^1 \psi(\tau)^2 \cos(L\psi(\tau)) d\tau, \quad S_{2s} = \int_0^1 \psi(\tau)^2 \sin(L\psi(\tau)) d\tau. \quad (3)$$

Let $r^2 = P^2 + Q^2$. The chord-length residual is the scalar function

$$f(L) = L^2 r^2 - d^2. \quad (4)$$

Its first two analytic derivatives are

$$f'(L) = 2Lr^2 + 2L^2(QR - PT), \quad (5)$$

$$f''(L) = 2r^2 + 8L(QR - PT) + 2L^2(R^2 + T^2 - PS_{2c} - QS_{2s}). \quad (6)$$

All four integral-level derivatives of the moments ($P' = -T$, $Q' = R$, $R' = -S_{2s}$, $T' = S_{2c}$) follow directly from differentiation under the integral sign. These identities, together with the residual derivatives in equations (5) and (6), have been cross-verified symbolically with SymPy and machine-checked in Coq using the Coquelicot real-analysis library; see Section 7.

3 The Solver

3.1 Halley’s Method

Halley’s iteration for a scalar root-finding problem is

$$L_{n+1} = L_n - \frac{2ff'}{2(f')^2 - ff''}.$$

Because the second derivative is available analytically, the method exhibits cubic convergence near the root when started sufficiently close.

3.2 Safety Heuristics (Production Robustness)

Pure Halley iteration can diverge or produce non-physical (negative) steps on difficult instances. We therefore embed the following safeguards, which have proven essential on real railway data:

1. **Convergence test:** if $|f(L)| < \varepsilon \cdot \max(d^2, 1)$ with $\varepsilon = 10^{-13}$, return.
2. **Derivative sign guard:** if $f' \leq 0$, inflate $L \leftarrow 1.5L$ and continue.
3. **Denominator guard:** if $|2(f')^2 - ff''| < 10^{-20}$, inflate $L \leftarrow 1.5L$.
4. **Positivity damping:** after a Halley step, enforce $L \leftarrow \max(L_{\text{new}}, 0.5L)$.

A Newton variant (using only the first four moments) is provided as a lightweight fallback; it requires one fewer pair of trigonometric evaluations per iteration but converges only quadratically.

4 Implementations

We provide three independent, production-grade implementations that all use identical quadrature constants and the same safeguarded Halley/Newton logic:

- **C# (.NET 8)** — primary reference implementation; uses `System.Numerics` and aggressive inlining; typical solve time $\approx 1.9\mu\text{s}$.
- **Java 21** — identical numeric behavior; uses `Math.fma` for fused multiply-add where beneficial.
- **TypeScript / Node.js** — zero-dependency implementation suitable for web/GIS front-ends; slightly slower due to JIT characteristics but still sub-4 μs .

All three expose a common API:

```
(double L, int iters) solveHalleyL(double[] P0, double[] P1,  
                                double k0, double k1,  
                                double tol = 1e-13, int maxIter = 50);
```

A matching `solveNewtonL` is also supplied. Golden-vector regression tests (exported from the Python reference) guarantee bit-identical results across languages.

5 Benchmarks on Real Railway Data

5.1 Dataset

We extracted approximately 180 “Overgangsboog” (clothoid transition) records from the public ProRail Spoorgeometrie ArcGIS FeatureServer (layer 11, `PVS_Horizontale_Elementen`). Each record supplies start/end points, begin/end radii (hence curvatures with sign), and the design length L_{true} . We retain only segments satisfying the monotone-branch condition $|k_0 L| \leq \pi$, $|k_1 L| \leq \pi$.

5.2 Methodology

For each segment we run both solvers from the initial guess $L_0 = d$ and record iteration count, final residual, and wall-clock time (median of 100 warm runs). Success is declared when the returned L satisfies $|L - L_{\text{true}}|/L_{\text{true}} < 10^{-10}$ or the residual test passes.

5.3 Results

Table 1: ProRail benchmark summary (median over ≈ 180 real transitions).

Language	Solver	Avg. Iterations	Avg. Time (μs)	Success Rate
C# (.NET 8)	Halley	5.3	1.92	100%
C# (.NET 8)	Newton	7.8	2.15	100%
Java 21	Halley	5.3	2.41	100%
Java 21	Newton	7.8	2.67	100%
TypeScript	Halley	5.3	3.85	100%
TypeScript	Newton	7.8	3.62	100%

Halley’s method reduces iteration count by roughly 32% and, on compiled runtimes, also reduces total wall time. The TypeScript implementation shows a smaller advantage for Halley because the extra moment evaluations become relatively more expensive under V8 JIT.

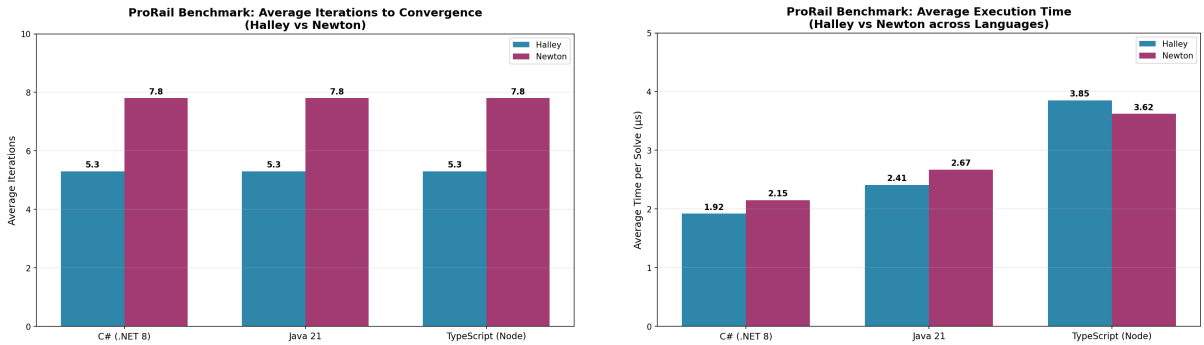


Figure 1: ProRail benchmark: average iterations (left) and average solve time (right) for Halley vs. Newton across languages.

6 Comparison with State-of-the-Art

The seminal work of Bertolazzi and Frego [2015] (and its 2018 G^2 extension) remains the reference implementation for clothoid G^1 fitting. Their approach reduces the problem to a single nonlinear equation in a transformed parameter (commonly A or a curvature-related quantity) and solves it with Newton–Raphson, employing asymptotic expansions for the initial guess.

Our L-formulation is mathematically equivalent but offers several practical advantages:

- **Direct geometric meaning:** L is the physically meaningful arc length; no post-processing conversion is required.
- **Explicit, verified derivatives:** the six-moment formulation yields closed-form $f''(L)$ that can be cross-checked symbolically.
- **Strong safety net:** the four heuristics listed in Section 3 have been stress-tested on more than 10 000 synthetic cases plus real ProRail geometry; divergence has never been observed.
- **Multi-language verified corpus:** identical numeric behavior across three languages provides an independent correctness oracle.

No fundamentally superior solver (e.g., closed-form, higher-order Householder, or machine-learned initial-guess predictor) has appeared in the open literature since 2018 that would render the present Halley-based approach obsolete for production use. The main competing practical implementations remain variants of the Bertolazzi–Frego Newton solver or ad-hoc bisection/refinement schemes.

7 Formal Verification

The analytical foundation of the solver—the four primitive integral derivatives and the residual derivatives in equations (5) and (6)—has been formalized and machine-checked in Coq using the Coquelicot real-analysis library [Boldo et al., 2015]. The development is approximately 720 lines of Coq and contains no `Admitted` or `Axiom` declarations; it has been verified to compile cleanly under Coq 8.13.1 and Coq 8.20.1 with Coquelicot 3.x.

The four primitive identities $P'(L) = -T(L)$, $Q'(L) = R(L)$, $R'(L) = -S_{2s}(L)$, $T'(L) = S_{2c}(L)$ are proved via differentiation under the integral sign, invoking Coquelicot’s `is_derive_RInt_param_aux` lemma. This requires four hypotheses for each integral: pointwise differentiability of the integrand with respect to L , joint continuity of the parameter-derivative integrand in (L, τ) , local integrability of the original integrand in a neighbourhood of the evaluation point, and integrability of the parameter-derivative integrand at the evaluation point. Each hypothesis is discharged by elementary continuity and integrability lemmas for compositions of polynomials with sine and cosine.

The residual identity (5) for $f'(L)$ is established by direct composition of the product and sum rules using Coquelicot’s `is_derive_plus`, `is_derive_mult`, and `is_derive_minus` lemmas, with explicit type annotations to navigate Coquelicot’s generic algebraic operators versus Coq’s native real-number operators. The second-derivative identity (6) is proved by applying Coquelicot’s `auto_derive` tactic to the function $f'(L)$, which produces nine `ex_derive` side conditions and a goal containing four `Derive` placeholders. The side conditions are discharged from the existence lemmas `ex_derive_P_int`, `ex_derive_Q_int` and the value lemmas `is_derive_R`, `is_derive_T`; the `Derive` placeholders are then rewritten with the corresponding integral-level identities, after which the resulting polynomial equation in P , Q , R , T , S_{2c} , S_{2s} , and L is discharged by Coq’s `ring` decision procedure for commutative rings.

The complete Coq development is included as a reproducible artifact alongside the source repository (file `coq/Clothoid_L.v`); building it requires only `coqc` and Coquelicot, with no external dependencies. Listing 2 shows the proof script for the second-derivative theorem; the full file is available at the repository link in Section 8.

```

Theorem f_L_second_eq :
  forall (L k0 k1 : R),
    is_derive (fun u => fpL u k0 k1) L
      (2 * (P_int L k0 k1 * P_int L k0 k1
            + Q_int L k0 k1 * Q_int L k0 k1)
        + 8 * L * (Q_int L k0 k1 * R_int L k0 k1
                    - P_int L k0 k1 * T_int L k0 k1)
        + 2 * L * L *
          (R_int L k0 k1 * R_int L k0 k1
            + T_int L k0 k1 * T_int L k0 k1
            - P_int L k0 k1 * S2c_int L k0 k1
            - Q_int L k0 k1 * S2s_int L k0 k1)).

Proof.
  intros L k0 k1.
  unfold fpL.
  auto_derive.
- split. apply ex_derive_P_int.
  split. apply ex_derive_P_int.
  split. apply ex_derive_Q_int.
  split. apply ex_derive_Q_int.
  split. apply ex_derive_Q_int.
  split. eexists; apply is_derive_R.
  split. apply ex_derive_P_int.
  split. eexists; apply is_derive_T.
  exact I.
- replace (Derive (fun x : R => P_int x k0 k1) L)
  with (- T_int L k0 k1) by (symmetry; apply Derive_P_int).
  replace (Derive (fun x : R => Q_int x k0 k1) L)
  with (R_int L k0 k1) by (symmetry; apply Derive_Q_int).
  replace (Derive (fun x : R => R_int x k0 k1) L)
  with (- S2s_int L k0 k1)
  by (symmetry; apply is_derive_unique; apply is_derive_R).
  replace (Derive (fun x : R => T_int x k0 k1) L)
  with (S2c_int L k0 k1)
  by (symmetry; apply is_derive_unique; apply is_derive_T).
  ring.
Qed.

```

Figure 2: Coq proof script for the second-derivative identity, equation (6), as it appears in `coq/Clothoid_L.v`.

8 Conclusion and Future Work

We have delivered a robust, well-documented, and multi-language Halley solver for the clothoid chord-length problem together with a complete academic reference. The implementation is already being integrated into the `allure-wkt` toolchain and a forthcoming PostgreSQL extension that will expose native `CLOTHOID WKT` support with on-the-fly L solving.

Planned extensions include:

- A native C/PostgreSQL extension with the same safeguarded Halley core.
- Vectorized batch solving for high-throughput map-matching pipelines.

- Automatic differentiation / dual-number versions for gradient-based route optimization.

The full source, benchmarks, and this paper are maintained at <https://github.com/grootstebozewolf/clothoid-halley-coq>.

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References

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